From Data to Detection:
Analysing Money Laundering Typologies with SQL and Power BI

Exploratory analysis of 9.5M synthetic transactions and three custom SQL detection rules, validated against labelled data.

9.5M	17	3	36.9%
transactions	typologies	SQL rules	laundering detected

Risk overview → | Typologies → | Schemes → | Detection rules →

Adam Płonkiewicz · PostgreSQL · Power BI

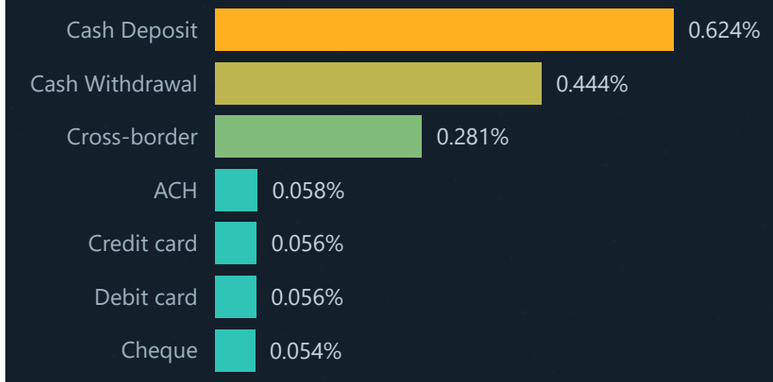
Source: SAML-D (Oztas et al., 2023) · CC BY-NC-SA 4.0

Money Laundering in the SAML-D synthetic dataset

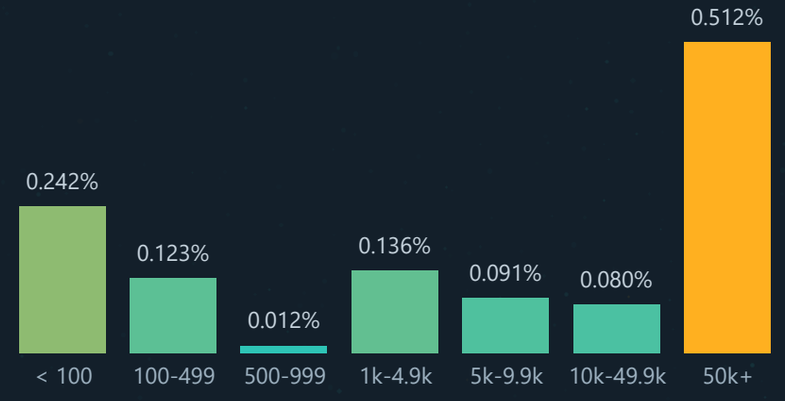
Laundering concentrates in cash and cross-border flows: cash deposits and payments to Nigeria or Morocco carry 6–8× the overall rate.

9,504,852	9,873	0.104%	0.481%
Total Transactions	Laundering Transactions	Laundering Rate	Laundering Amount Share

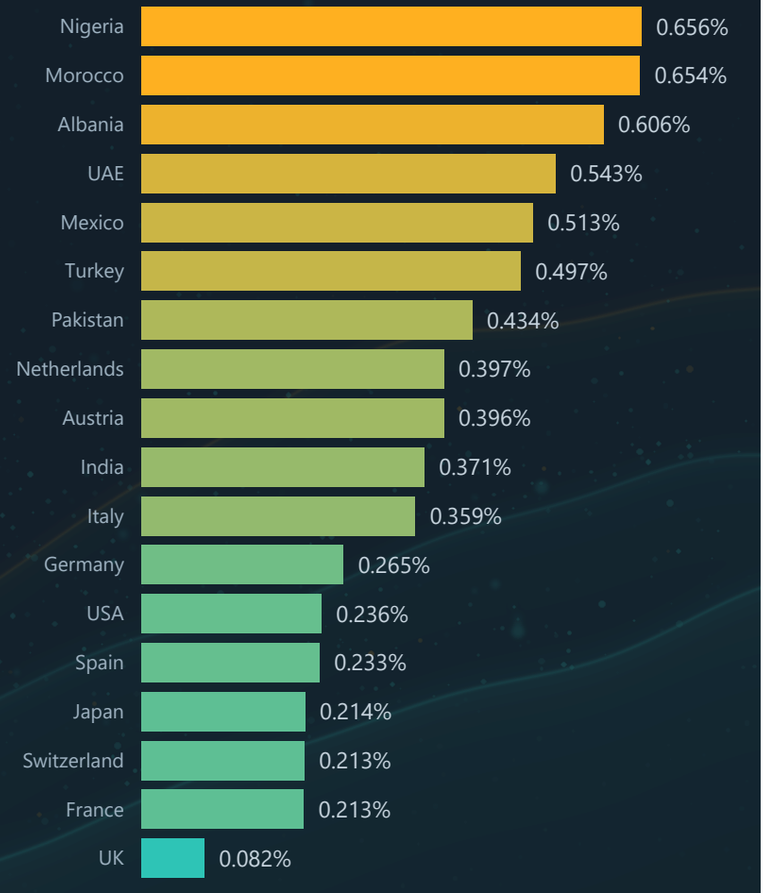
Laundering rate by payment type



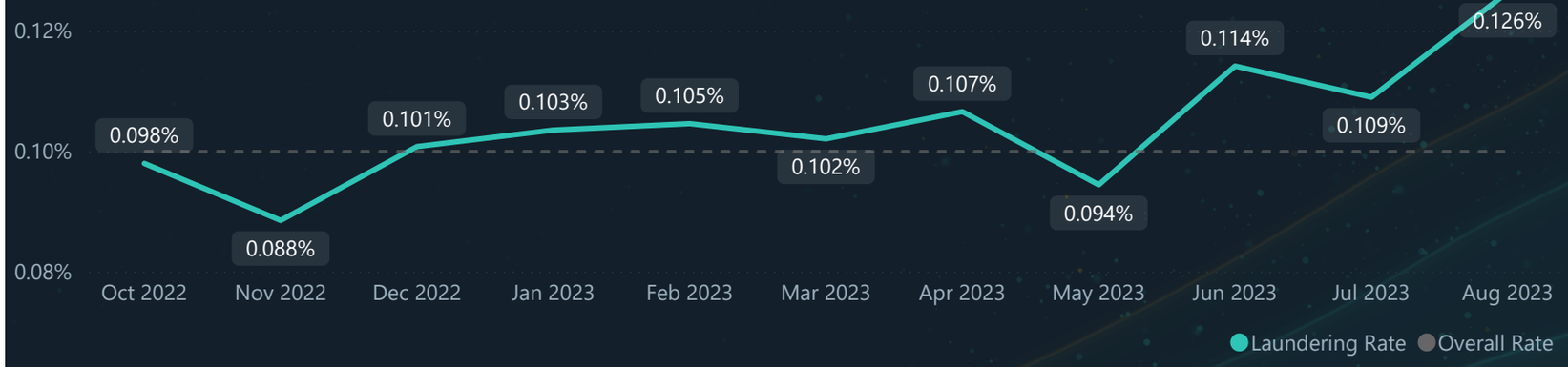
Laundering rate by transaction amount (GBP)



Laundering rate by receiver bank country



Laundering rate over time



Source: SAML-D (Oztas et al., 2023), synthetic data · CC BY-NC-SA 4.0

Money Laundering in the SAML-D synthetic dataset

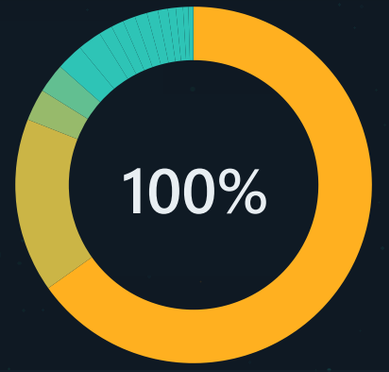
Over-Invoicing makes up 65% of laundering value – likely by design: it is the only typology above £1M, and no legitimate transaction in the dataset exceeds that amount.

9,873	12,618,500	16
Laundering transactions	Max Amount (GBP)	Min Amount (GBP)

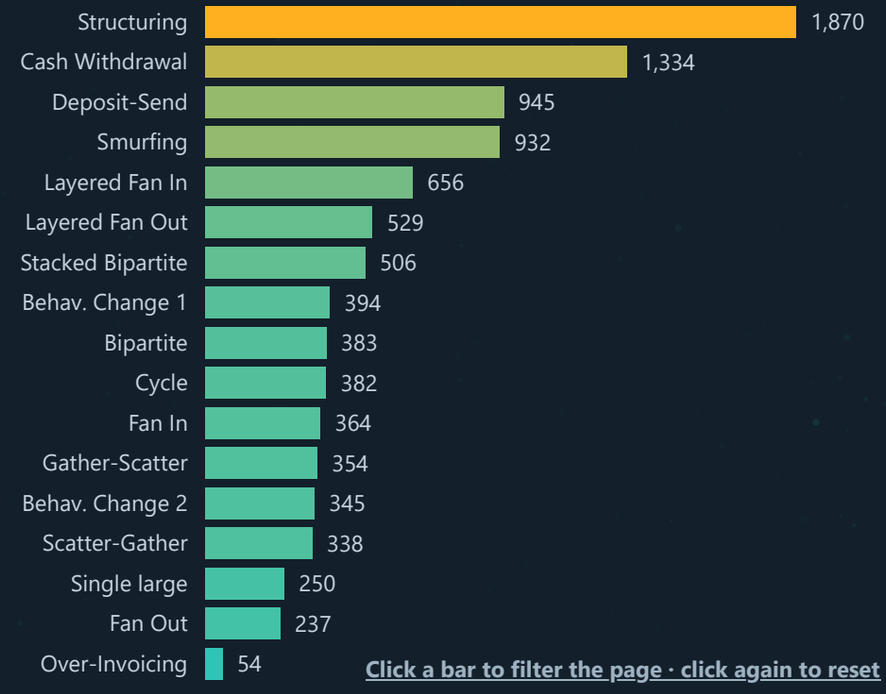
401M	5,323
Laundering amount (GBP)	Median amount (GBP)

All typologies

share of laundered money (GBP)



Laundering transactions by typology



Click a bar to filter the page · click again to reset

Payment types used by each typology

Laundering Type	ACH	Cash Deposit	Cash Withdrawal	Cheque	Credit card	Cross-border	Debit card
Behav. Change 1	11%			11%	7%	65%	6%
Behav. Change 2						100%	
Bipartite	20%			19%	19%	24%	19%
Cash Withdrawal			100%				
Cycle	16%			13%	16%	42%	12%
Deposit-Send	3%	50%		4%	4%	36%	4%
Fan In	22%			20%	23%	13%	23%
Fan Out	16%			17%	16%	38%	14%
Gather-Scatter	21%			18%	21%	14%	26%
Layered Fan In	22%			19%	19%	20%	20%
Layered Fan Out	19%			17%	21%	25%	18%
Over-Invoicing	19%			19%	15%	11%	37%
Scatter-Gather	17%			14%	12%	38%	18%
Single large	23%			19%	21%	21%	16%
Smurfing		100%					
Stacked Bipartite	20%			20%	21%	15%	23%
Structuring	15%			16%	16%	39%	15%

Source: SAML-D (Oztas et al., 2023), synthetic data · CC BY-NC-SA 4.0

Money Laundering in the SAML-D synthetic dataset

1. **Cash withdrawals:** laundered money is taken out of the account as cash, in a series of small withdrawals that look ordinary on their own. Here, after weeks without any withdrawal, the laundering account withdraws cash every day for five days, while the legitimate account withdraws steadily every few days (13 Jul – 23 Aug 2023).

Laundering account

Number of cash withdrawals per day



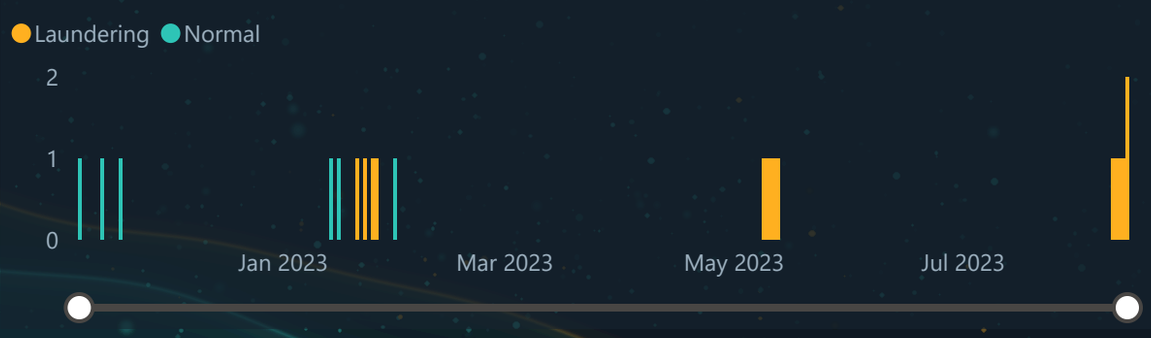
Legitimate account

Number of cash withdrawals per day



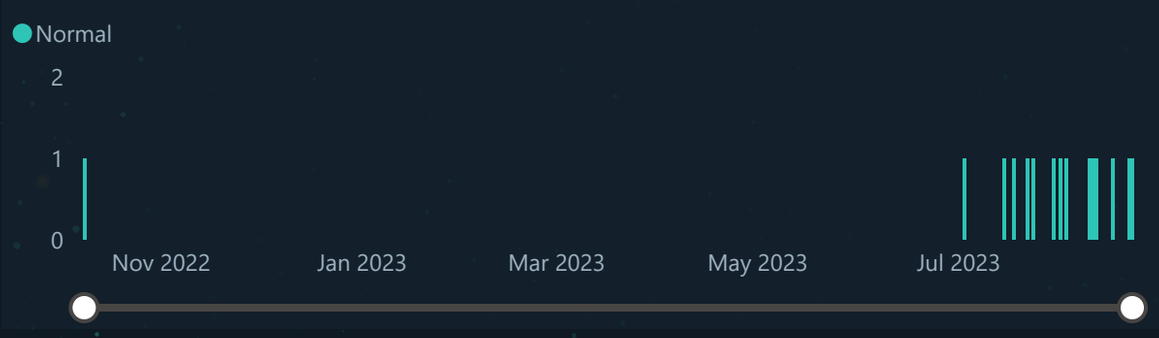
Laundering account

Number of cash withdrawals per day – full period (drag to zoom)



Legitimate account

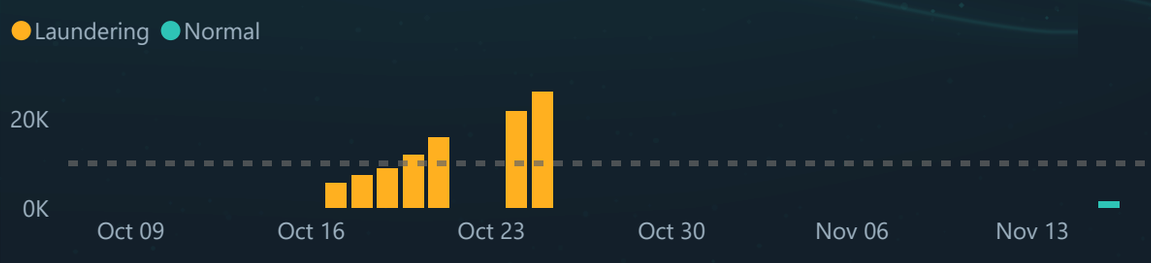
Number of cash withdrawals per day – full period (drag to zoom)



2. **Structuring:** one person splits a large amount of cash into several smaller deposits, so that no single deposit reaches the reporting threshold (assumed £10k). Here, every deposit stays well below £10k, yet together they exceed it within just ten days (7 Oct – 17 Nov 2022).

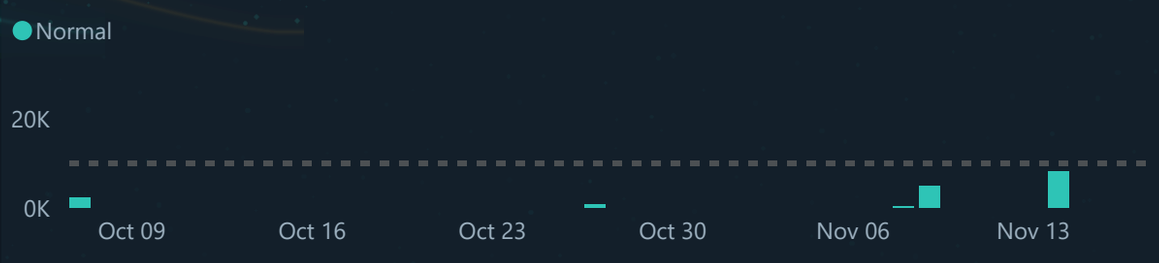
Laundering account

10-day total of cash deposits from the same depositor (GBP)



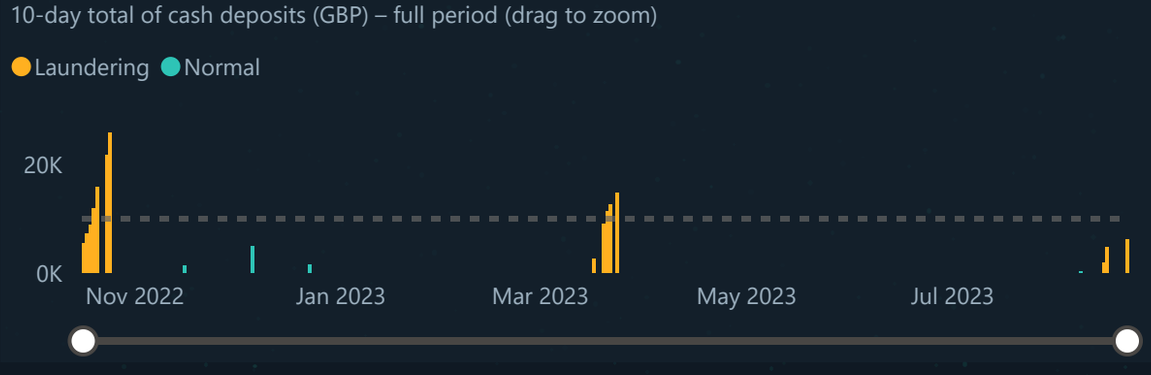
Legitimate account

10-day total of cash deposits from the same depositor (GBP)



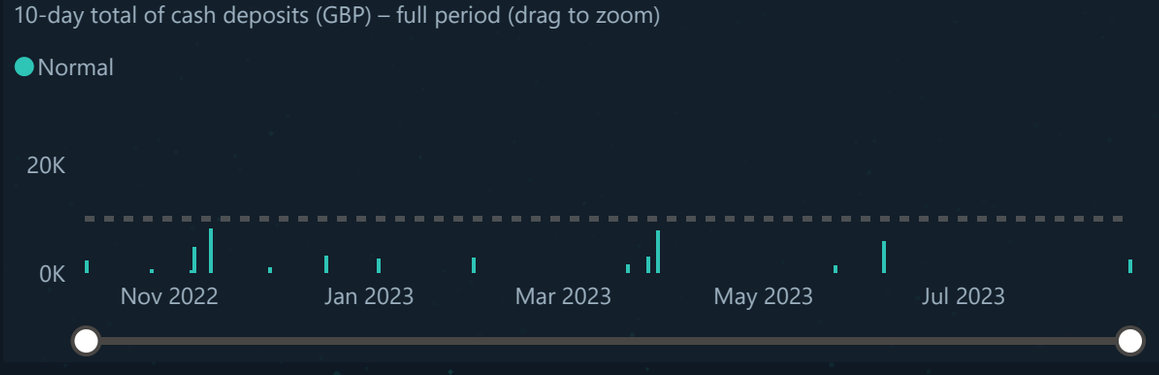
Laundering account

10-day total of cash deposits (GBP) – full period (drag to zoom)



Legitimate account

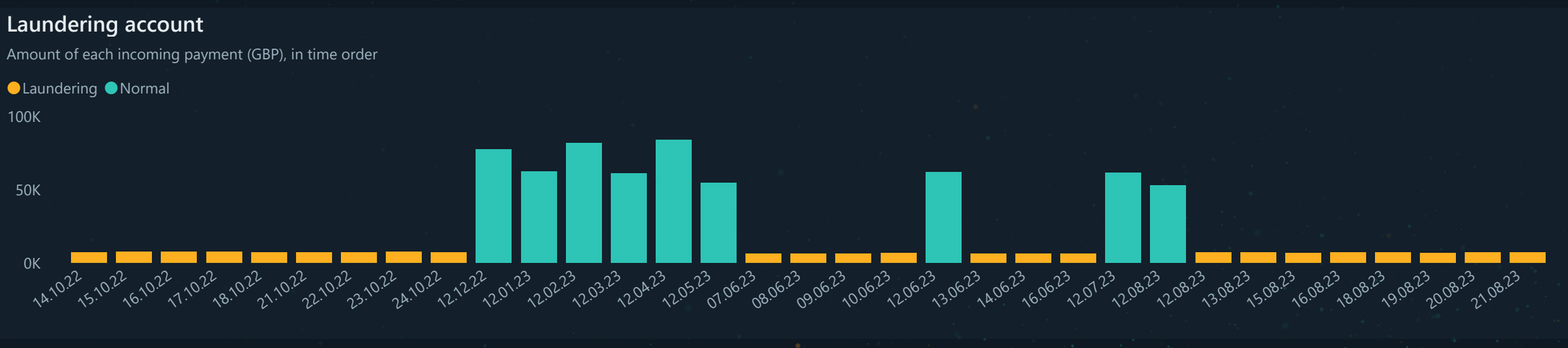
10-day total of cash deposits (GBP) – full period (drag to zoom)



3. **Smurfing:** many people ("smurfs") each send a similar amount to one collecting account, so that the money arrives in small, unremarkable pieces. On the laundering account, payments come in short series from different first-time senders, all for almost the same amount; a legitimate account receives payments from regular senders, each with its own typical amount.

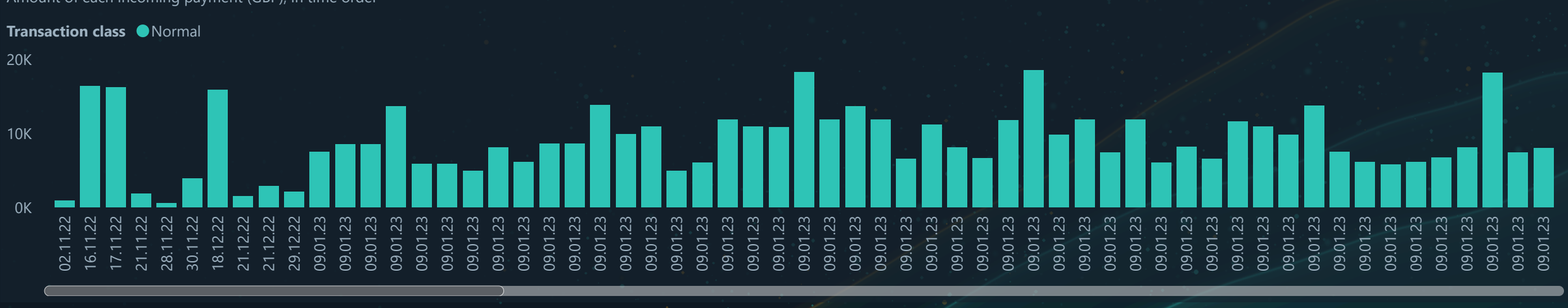
Laundering account

Amount of each incoming payment (GBP), in time order



Legitimate account

Amount of each incoming payment (GBP), in time order



In all three schemes, a single laundering transaction looks ordinary – it is the pattern over time that gives it away: a sudden burst, a running total, or a series of near-identical payments.

Source: SAML-D (Oztas et al., 2023), synthetic data · CC BY-NC-SA 4.0

Detection rules

How well do simple SQL rules detect the three schemes?

Targeted typologies

41.9%
of laundering transactions

3.2%
of laundering value

Detected by rules

36.9%
of laundering transactions

3.0%
of laundering value

The rules catch 88% of transactions from the three targeted schemes (36.9% of 41.9%), but these schemes carry only about 3% of laundering value – most of the money moves through typologies the rules do not cover.

Structuring rule - detects SAML-D label "Smurfing"

Flags 5 or more cash deposits from the same depositor to the same account within 10 days, each below £10k, totalling at least £10k.

97.8% precision · 99.2% recall

932	→	925	7	21	946
laundering		detected	missed	false alarms	flagged

Smurfing rule - *detects SAML-D label "Structuring"*

Flags 5 or more payments from first-time senders to the same account within 10 days, with amounts within 10% of each other.

98.9% precision · 93.2% recall

1,870	→	1,742	128	20	1,762
laundering		detected	missed	false alarms	flagged

Cash withdrawal rule - *detects SAML-D label "Cash Withdrawal"*

Flags 6 or more cash withdrawals within 7 days by an account that made no withdrawals in the previous 30 days.

86.7% precision · 73.1% recall

1,334	→	975	359	150	1,125
laundering		detected	missed	false alarms	flagged

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